

1 Conjugate Prior Distributions

Let $\{f(x|\theta) : \theta \in \Theta\}$ be a family of distributions of interest. A class Π of probability distributions forms a **conjugate family** if the posterior density

$$\pi(\theta|x) \propto \pi(\theta) f(x|\theta).$$

is in Π for all x and all $\pi \in \Pi$.

NOTE: The conjugate family should be defined on the entire parameter space Θ . If for $\Theta_1 \subsetneq \Theta$, $\pi(\theta) = 0$, the posterior can **never** have positive probabilities on $x \in \Theta_1$ even if $f(x|\theta) > 0$.

2 Bernoulli Distribution

Suppose the distribution of interest

$$\mathbf{X} \sim \text{Bern}(\theta), \quad f(x|\theta) = \theta^x (1-\theta)^{1-x}.$$

The conjugate family for θ is the **Beta distribution**. Suppose $\theta \sim \text{Beta}(\alpha, \beta)$, where $\alpha, \beta > 0$:

$$\pi(\theta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}, \quad 0 < \theta < 1.$$

The posterior distribution is

$$\pi(\theta|\mathbf{x}) = \text{Beta}(\alpha + n\bar{x}, \beta + n - n\bar{x}).$$

3 Poisson Distribution

Suppose the distribution of interest

$$\mathbf{X} \sim \text{Poisson}(\lambda), \quad f(x|\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

The conjugate family for θ is the **Gamma distribution**. Suppose

$\theta \sim \text{Gamma}\left(\alpha, \frac{1}{\beta}\right)$, where $\alpha, \beta > 0$:

$$\pi(\lambda) = \lambda^{\alpha-1} e^{(-\beta\lambda)}.$$

The posterior distribution is

$$\pi(\lambda|\mathbf{x}) = \text{Gamma}\left(\alpha + \sum_{i=1}^n x_i, \frac{1}{\beta + n}\right).$$

4 Exponential Distribution

Suppose the distribution of interest

$$\mathbf{X} \sim \text{Exponential}\left(\frac{1}{\lambda}\right), \quad f(x|\lambda) = \lambda e^{-\lambda x} \quad \lambda, x > 0.$$

The conjugate family for λ is the **Gamma distribution**. Suppose

$\theta \sim \text{Gamma}\left(\alpha, \frac{1}{\beta}\right)$, where $\alpha, \beta > 0$:

$$\pi(\lambda) = \lambda^{\alpha-1} e^{(-\beta\lambda)}.$$

The posterior distribution is

$$\pi(\lambda|\mathbf{x}) = \text{Gamma}\left(\alpha + n, \frac{1}{\beta + \sum_{i=1}^n x_i}\right).$$